

LAC BENOIT, QC, Canada

WMO: 715200

Lat: 51.532N

Lon: 71.111W

Elev: 549

StdP: 94.90

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.5	-38.1	-45.0	0.0	-41.0	-41.6	0.1	-38.0	12.4	-12.3	10.8	-12.6	0.6	150	0.788

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	25.9	16.8	23.9	15.9	22.0	15.4	19.0	22.4	17.8	21.2	16.7	19.9	4.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.7	20.0	16.7	12.7	18.9	15.5	11.8	17.9	56.3	22.3	52.6	21.3	49.2	19.9	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.4	8.4	DB	-46.5	29.7	2.3	1.7	-48.1	31.0	-49.5	31.9	-50.8	32.9	-52.5	34.1	(4)
(5)			WB	-45.7	21.1	1.7	1.0	-46.9	21.8	-47.9	22.4	-48.9	22.9	-50.2	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	-21.2	-20.3	-12.6	-3.2	5.1	11.8	14.6	13.5	8.5	1.6	-6.7	-16.4
	DBStd	14.01	8.43	8.32	8.52	5.96	4.47	4.34	3.14	3.29	4.07	4.25	6.02	7.99
	HDD10.0	4767	967	848	701	394	163	29	2	7	77	262	500	817
	HDD18.3	7440	1226	1081	959	644	412	201	122	154	297	519	750	1075
	CDD10.0	384	0	0	0	0	11	83	144	114	31	1	0	0
	CDD18.3	14	0	0	0	0	0	4	6	3	1	0	0	0
	CDH23.3	225	0	0	0	0	10	71	94	44	6	0	0	0
	CDH26.7	31	0	0	0	0	1	13	13	4	1	0	0	0
Wind	WSAvg	3.7	3.6	3.3	3.6	3.8	3.5	3.6	3.7	3.7	4.1	4.2	4.1	3.4
Precipitation	PrecAvg	922	48	40	47	60	70	91	124	117	108	82	77	57
	PrecMax	1082	91	70	84	102	146	152	180	175	230	119	119	106
	PrecMin	760	24	12	14	18	35	37	66	59	49	49	46	30
	PrecStd	82	19	13	17	20	23	29	27	31	35	21	20	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	2.2	7.4	14.0	24.7	28.4	28.4	27.1	23.7	15.6	7.1	1.8
		MCWB	0.7	0.9	3.1	5.9	13.3	18.2	18.2	18.3	17.1	11.4	6.4	1.3
	2%	DB	-1.4	-0.4	4.2	10.3	20.3	25.3	25.9	24.4	20.5	12.3	4.2	0.2
		MCWB	-1.7	-1.6	1.1	4.4	11.8	15.9	17.1	17.0	15.5	9.9	3.8	0.0
	5%	DB	-5.3	-3.6	2.0	7.7	17.2	23.0	23.9	22.1	17.9	10.2	2.1	-2.2
		MCWB	-5.6	-4.5	-0.2	3.1	10.2	15.0	16.2	16.3	14.6	8.7	1.7	-2.5
	10%	DB	-8.1	-6.6	0.2	5.5	14.2	20.8	21.9	20.1	15.6	8.2	0.8	-4.5
		MCWB	-8.5	-7.4	-1.3	1.8	8.7	13.7	15.6	15.4	13.0	6.9	0.5	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.0	4.2	6.7	15.2	20.0	20.5	20.2	18.6	12.9	6.6	1.5
		MCDB	0.6	1.9	6.0	12.1	20.3	24.5	24.5	23.5	20.7	14.3	6.7	1.7
	2%	WB	-1.8	-1.5	1.9	4.9	13.2	18.1	18.9	18.7	16.7	10.7	3.6	0.1
		MCDB	-1.6	-0.4	3.5	9.2	18.1	22.3	22.4	21.9	19.3	11.8	3.8	0.3
	5%	WB	-5.6	-4.4	0.2	3.7	11.4	16.3	17.7	17.5	15.1	9.1	1.6	-2.4
		MCDB	-5.3	-3.5	1.7	6.8	16.1	20.4	21.5	20.2	17.1	10.3	1.8	-2.2
	10%	WB	-8.4	-7.6	-1.4	2.5	9.5	14.9	16.7	16.4	13.4	7.1	0.4	-4.6
		MCDB	-8.1	-6.8	0.4	5.1	14.1	19.0	20.2	18.9	15.3	8.1	0.6	-4.4
Mean Daily Temperature Range	5% DB	MDBR	13.4	16.0	16.4	13.5	12.8	13.7	12.5	11.6	10.4	7.4	7.6	11.2
		MCDBR	11.4	12.4	15.0	15.8	19.1	18.0	16.7	15.1	14.2	10.9	6.7	8.4
		MCWBR	11.1	11.4	12.1	10.6	11.4	10.0	9.0	8.8	9.3	8.6	6.3	7.9
	5% WB	MCDBR	11.3	11.9	13.2	12.9	16.4	13.8	12.6	11.9	11.3	9.7	6.5	8.2
		MCWBR	11.2	10.9	10.8	8.9	10.6	8.5	7.7	7.8	8.3	8.1	6.2	8.2
Clear-Sky Solar Irradiance	taub	0.245	0.259	0.272	0.317	0.346	0.364	0.377	0.353	0.335	0.302	0.270	0.243	
	taud	2.222	2.220	2.224	2.186	2.375	2.395	2.383	2.469	2.491	2.532	2.369	2.211	
	Ebn at Noon	780	874	931	920	907	890	871	877	855	825	749	712	
	Edh at Noon	67	90	109	130	114	113	112	98	86	67	59	57	
All-Sky Solar Radiation	RadAvg	1.18	2.33	3.76	4.92	5.44	5.38	4.81	3.85	2.92	1.62	0.80	0.71	
	RadStd	0.10	0.17	0.29	0.36	0.35	0.44	0.41	0.26	0.22	0.16	0.12	0.11	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page